

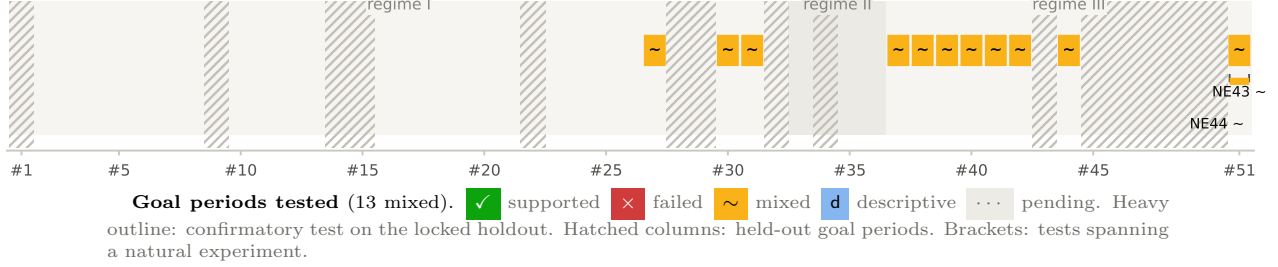
H16: Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2/5 Usefulness 2.5/5

Bottom line. Pause-chain traps are held by the context: a forced erasure ends most of them, and a message works by naming the agent, not by diluting its repeats.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

The units are per-agent trap spells: inactive spells and timer-pause chains, on the wall clock (30-s bins) and on the call clock (idle gates of the context ledger). We test them in 11 non-reserved goal periods. Kramers escape predicts a constant hazard; traps *age* instead (#51 $\beta = -0.77$). Round 2 asks why. A Pólya urn says the context fills with the agent's own repeats. It predicts the gate-clock exponent from composition alone (-0.36 vs -0.35) and absorbs half of it, but not the wall-clock one (-0.08 vs -0.77). The decisive test is the forced erasure at the 41-call cap (NE41). It lifts sustained escape at the next gate from 0.47 to 0.84 (log OR +2.68 [2.31, 3.16]), 4.6× what dilution implies. A per-trap frailty world gives no step. A directed message works by address, not dilution. Trap depth is not a model trait. Spell-kind mixing explains little of #51's aging and most of it in two 4-h periods. The reserved data are not used.

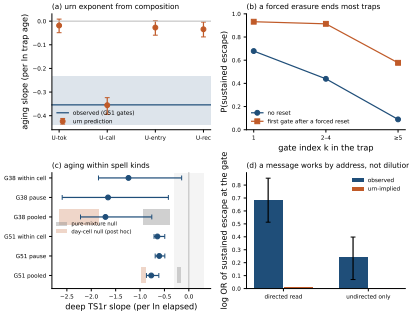
PHYSICAL MODEL

Model 02 with an urn on the context window. At an idle gate g (a call after a pause call) the agent acts or re-pauses. The context segment since the last reset holds n_{rep} own idle calls, n_{act} own active calls and k room items. The urn sets escape by the own-repeat share f , measured four ways: tokens, own calls, all entries, the last 10 entries. Rivals: aging with trap age a at fixed f ; a context step at erasure; a mixture of spell kinds; per-trap frailty. Fit: agent effects α_i , slopes β_a , β_f , reset step δ , read effects γ .

$$P(\text{escape}_g) = c_i (1 - f_g), \quad f_{\text{call}} = \frac{n_{\text{rep}}}{n_{\text{rep}} + n_{\text{act}}}$$

$$\text{logit } P_g = \alpha_i + \beta_a \ln a_g + \beta_f \ln(1 - f_g) + \delta \mathbb{I}[\text{reset}] + \gamma_d \mathbb{I}[\text{dir}] + \gamma_u \mathbb{I}[\text{undir}]$$

$$\text{urn: } \beta_f \approx 1, \beta_a \rightarrow 0, \gamma_d = \gamma_u \approx \Delta \ln(1 - f)$$



Observables. (a) #51 gate aging slope (blue band: observed, day-bootstrap CI) against the slope each urn ruler predicts from measured composition, with no fitted slope (orange). Only the own-call share matches. (b) Escape at the next gate after a forced erasure vs without one, by depth in the trap. (c) Wall-clock aging within spell kinds; grey and orange bands are the pure-mixture and day-cell nulls. (d) A directed read lifts escape about 3× more than an undirected one; the urn implies nearly nothing for both.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

When an agent re-pauses again and again, erase or consolidate its context. In #51 that ends 84% of traps at the next gate, against 47% otherwise, at any depth. Deep traps (5+ re-pauses) still escape less after the erasure (0.58 vs 0.93), so some depth lives outside the context. If you message a stuck agent, name it. A directed read adds +0.69 log odds, room chatter +0.24; a second message adds about a third. Do not expect a model to be reliably more trap-prone: depth does not repeat across periods.

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RESULTS VS PREDICTIONS

prediction (round 2)	observed (#51)	verdict
token urn misses the exponent	-0.02 predicted vs -0.35 holds	
own-call urn: gate yes, wall clock no	-0.36 vs -0.35 ; -0.08 vs -0.77 holds	
composition absorbs 25%	$\rho = 0.47$ [0.33, 0.70], $\beta_f = 1.05$ failed	
forced erasure raises escape	log OR +2.68 [2.31, 3.16] holds	
erasure dose	+0.11 [-0.35, 0.44] no power	
directed > undirected	+0.45 [0.32, 0.58] holds	
urn-implied kick < 0.2× observed	0.01 vs 0.69 holds	
depth invariant across periods	$r = -0.04$ [-0.55, 0.22] failed	
aging slopes differ by agent	$I^2 = 0.92$ holds	
pause spells age; beat mixture null	-0.61 ; null -0.20 holds	

GOAL PERIODS IN DETAIL

G27 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G30 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G31 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G37 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G38 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G39 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G40 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G41 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G42 ~ 1b: mixed (unchanged under the round-1 rule; its core...
G44 ~ 1b: mixed (unchanged under the round-1 rule; its core...
+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	2		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

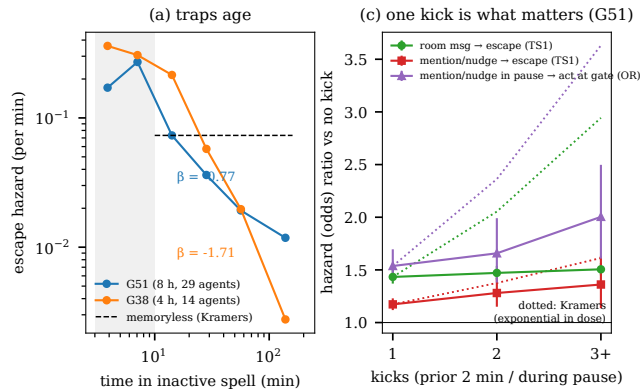
The reset and dose tests are powered only in #51. A day-level mixture of escape rates (post hoc) reproduces the wall-clock exponent (-0.93), so that exponent is not separable from day states; the gate results are not affected. With the gate count in the model (post hoc), trap age turns timer-like and the count carries the aging. Per-trap frailty fakes gate aging in synthetic worlds; only the erasure separates it. Kick exposure is now the ledger read at the gate; no in-flight placebo was run in round 2.

ROUND 2

Confirm the erasure step on reserved #51 days and #45. Test what survives the erasure (memory, task state) in deep traps.

- **Essence.** Traps deepen by self-reinforcement as the context fills with the agent's own output.
- **H16-R1** Pólya urn: $P(\text{repeat}) \propto$ the share of the context that is own repeats.
- **H16-R2** One directed message works because it dilutes the urn; repeated messages add little once the context is diluted.
- **H16-R3** Trap depth is a model trait: some models loop more.

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Second figure. Round 1. (a) Escape hazard from inactive spells vs time in the spell in the two periods with the most power; Kramers predicts the flat dashed line. (b) #51 hazard and odds ratios vs the number of kicks; the first message carries the effect.